CSA2001

FUNDAMENTALS OF AI AND ML EVALUATED COURSE PROJECT

PROJECT REPORT-AI Resume Screening System

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1. Introduction

Recruitment plays a critical role in every organization, yet it often becomes a major bottleneck. With companies receiving hundreds—or even thousands—of applications for a single position, manually reviewing each resume is both time-intensive and highly inefficient.

This project introduces an **AI-Powered Resume Screening System** designed to automate the initial evaluation process. The system intelligently parses resumes, extracts key information, and matches candidate profiles against a given job description to assess overall suitability.

Leveraging **Natural Language Processing (NLP)** techniques, it performs text extraction, identifies relevant skills and qualifications, and computes a similarity or matching score between the resume and the job requirements.

The primary objective of this project is to showcase how Artificial Intelligence can streamline resume screening, enabling recruiters to focus their time and effort on the most promising candidates and significantly accelerate the hiring process.

2. Problem Statement

Recruiters frequently encounter several challenges during the hiring process, including:

- Reviewing a massive volume of resumes manually
- Struggling to quickly spot relevant skills and qualifications
- Spending excessive time on initial candidate screening
- Risk of unintentional human bias or oversight

To overcome these issues, many organizations are now adopting Applicant Tracking Systems (ATS) and AI-powered tools that can automatically filter and rank resumes.

This project focuses on developing a **simple yet effective AI-based Resume Screening System**. The system analyzes uploaded resumes and compares them with a given job description to generate an automatic match score, helping recruiters identify the most suitable candidates efficiently.

3. Objective of the Project

The primary goals of this AI Resume Screening System are:

- To develop an automated system capable of intelligently analyzing resumes.
- To effectively compare candidate resumes with the provided job description using Natural Language Processing (NLP) techniques.
- To compute a reliable similarity or match score that indicates how well a resume aligns with the job requirements.
- To accurately extract key skills, qualifications, and experiences from resumes.

- To generate constructive feedback highlighting missing or required skills and overall candidate suitability for the role.

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4. Technologies Used

The project uses the following technologies:

Programming Language

- Python

Web Framework

- Flask

AI / NLP Techniques

- TF-IDF Vectorization
- Cosine Similarity

Libraries

- pdfplumber (PDF text extraction)
- python-docx (Word file extraction)
- scikit-learn (machine learning algorithms)
- sentence-transformers (AI embeddings)
- spaCy (NLP processing)

Frontend

- HTML
- CSS

Development Tools

- VS Code
- Git
- GitHub

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5. System Architecture

The working of the system follows the following pipeline:

User Uploads Resume

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Resume Text Extraction

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Skill Extraction

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TF-IDF Vectorization

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Cosine Similarity Calculation

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Match Score Generation

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Result Display (Score + Skills + Feedback)

The system processes both the **resume text and job description** and calculates a similarity score that indicates how well the resume matches the job.

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6. Features of the System

The AI Resume Screening System offers the following powerful features to streamline the recruitment process:

- **Resume Upload** Users can easily upload resumes in PDF or DOCX formats.
- **Resume Text Extraction** The system automatically extracts text from the uploaded resume using efficient Python libraries.
- **Resume–Job Description Matching** It intelligently compares the resume content with the provided job description using advanced Natural Language Processing (NLP) techniques.
- **Match Score Calculation** The system generates a percentage-based similarity score that indicates how well the candidate's profile matches the job requirements. *Example: Resume Match Score: 78%*
- **Skill Extraction** Key technical and professional skills are automatically identified from the resume. *Example: Python, Machine Learning, SQL, Flask*
- **Missing Skills Detection** The system highlights important skills required in the job description that are absent from the resume. *Example: Missing Skills: Docker, AWS*
- **Candidate Rating** Based on the overall match score, the system categorizes candidates into clear categories:
    - Excellent Candidate
    - Good Candidate
    - Average Candidate
    - Poor Match

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7. Implementation

The AI Resume Screening System was developed using **Python** and **Flask** as the primary technologies.

The backend handles resume processing, text extraction, skill identification, and NLP-based matching with the job description. The frontend provides an intuitive web interface that allows users to upload resumes and input job descriptions seamlessly.

Main Project Files and Structure

The project follows a clean and organized structure with the following key files and directories:

- **app.py** — Main Flask application file containing the server logic and route handling.
- **resume_parser.py** — Responsible for extracting text and identifying skills from uploaded resumes.
- **similarity.py** — Implements NLP techniques to calculate the similarity score between the resume and job description.
- **templates/** — Folder containing all HTML files for the user interface.
- **static/** — Directory for CSS stylesheets and other static assets.

The entire project is version-controlled and hosted on **GitHub**, making it easy to collaborate and track changes.

8. Challenges Faced

During the development of this project, several challenges were encountered:

- Extracting text from different resume formats
- Handling missing Python libraries
- Resolving Git and GitHub integration issues
- Improving the frontend design
- Ensuring accurate skill extraction

These challenges were addressed through debugging, research, and iterative improvements.

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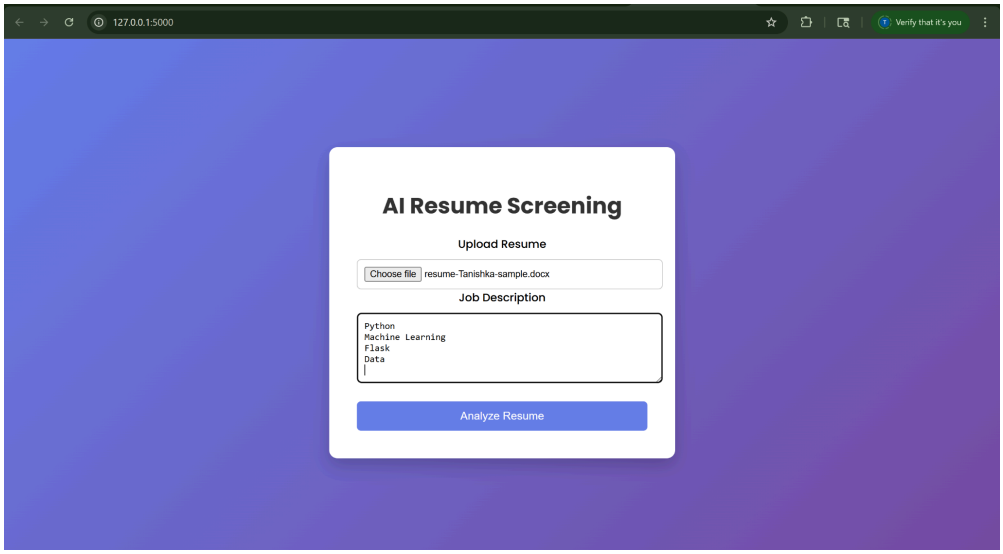
9. Results

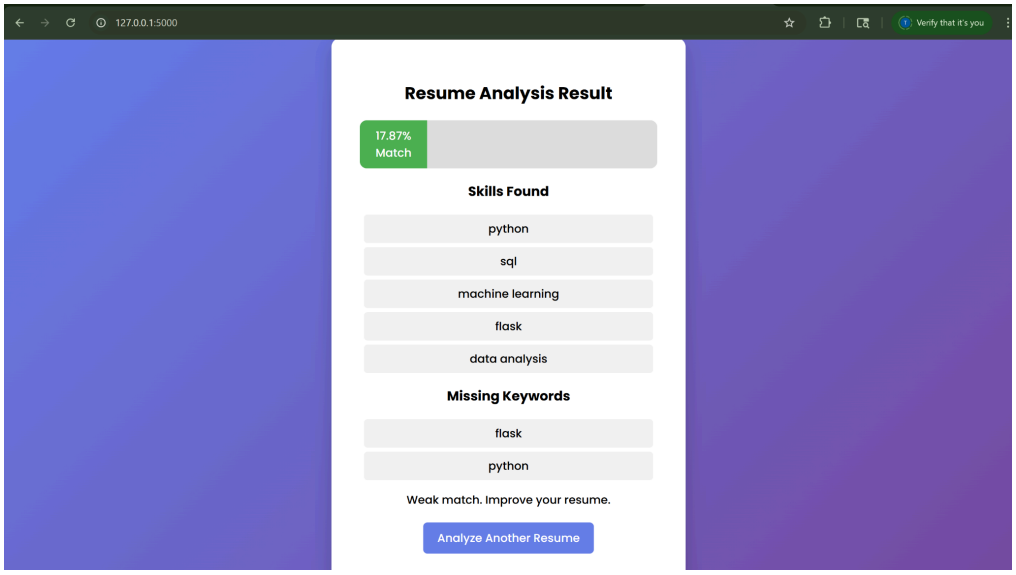
The system successfully analyzes resumes and provides:

- Resume similarity score
- Extracted skills
- Missing skills
- Candidate rating

This demonstrates how AI can assist recruiters by quickly identifying suitable candidates.

```
(venv) (base) PS C:\Users\Lenovo\OneDrive\Desktop\AI-Resume-Screening-System> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
```





10. Future Improvements

While the current version of the AI Resume Screening System delivers strong core functionality, several improvements can be implemented to make it more robust and closer to real-world recruitment platforms:

- **Support for Multiple Resume Uploads** Allowing users to upload and process multiple resumes simultaneously for faster batch screening.
- **Resume Ranking System** Automatically ranking all uploaded resumes in descending order based on their match scores.
- **Advanced AI Models** Integrating powerful transformer-based models such as BERT or other pre-trained language models to improve the accuracy of skill extraction and semantic matching.
- **Automatic Job Description Extraction** Enabling the system to intelligently parse and extract job requirements from uploaded job postings or company descriptions.
- **Enhanced User Interface and Experience (UI/UX)** Redesigning the frontend with a more modern, responsive, and user-friendly interface to improve usability.

These enhancements would significantly increase the system's efficiency, scalability, and practicality, bringing it closer to advanced AI-powered recruitment solutions used in the industry.

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11. Conclusion

The AI Resume Screening System effectively demonstrates the power of Artificial Intelligence and Natural Language Processing in automating one of the most time-consuming stages of recruitment.

By intelligently comparing resumes against job descriptions and accurately extracting relevant skills, the system significantly reduces manual effort and helps recruiters focus on high-potential candidates.

Although this project represents a simplified prototype, it clearly showcases the immense potential of AI to enhance hiring efficiency, minimize screening time, and bring greater objectivity to the recruitment process.

12. GitHub Repository

GitHub Link:

https://github.com/tanishkasahu191-ui/AI-Resume-Screening-System